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Techno-Economic Analysis of Off-Grid Hybrid Renewable Energy System for Ethiopian Rural Electrification

Abiy Mekonnen[1,*], Ravikumar Hiremath[2] and Dereje Shiferaw[3]

1 Department of Electrical and Computer Engineering, Addis Ababa Science and Technology University, Addis Ababa, Ethiopia
2 Department of Electrical and Computer Engineering, Adama Science and Technology University, Adama, Ethiopia
3 School of Electrical and Computer Engineering, Addis Ababa Institute of Technology, Addis Ababa, Ethiopia
*Corresponding author. E-mail: abiy.mekonnen@aastu.edu.et

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Abstract

This study presents a comprehensive plan for implementing off-grid hybrid renewable power systems in rural areas of Ethiopia, as a part of the government's ambitious renewable energy development initiatives. The focus is on leveraging the country's abundant solar, wind, and micro-hydro power resources through careful planning and analysis. Through field surveys, data collection on local population density, electricity demand, and available renewable energy potential, this study identifies key factors for designing effective off-grid hybrid systems. Utilizing NASA's wind speed and solar radiation databases, alongside on-site measurements and government data on micro-hydro potential, the study modeled three different hybrid systems using HOMER software. The analysis considers various factors such as net present cost, capital investments, and energy costs to determine the feasibility of each hybrid system. For efficient solar and wind resource scenarios, the cost of energy can be as low as $0.122/kWh, while optimal solar radiation and micro-hydro conditions can lower it to $0.043/kWh. In cases where renewable resources may be limited, hybridizing with diesel generators can provide a cost-effective solution, albeit at a slightly higher cost of $0.14/kWh. Taking into account the sensitivity analyses of Photovoltaic (PV) and diesel prices, this study demonstrates that the proposed off-grid hybrid renewable systems can offer competitive energy costs compared to the current national tariff. With a focus on 180 rural communities within a mile radius, this holistic approach to renewable energy planning holds significant promise for sustainable and affordable power supply in Ethiopia's rural areas.

Keywords: hybrid renewable, load forecasting, planning, resource estimation

1. Introduction

Renewable energy has emerged as a promising solution to address the pressing concerns related to climate change and environmental degradation [1, 2]. In a world where the consequences of climate change are becoming increasingly evident, transitioning to renewable energy sources is not optional but a necessity for ensuring a sustainable future for generations to come [3]. Various renewable energy sources such as solar, wind, biomass, and micro-hydro energy have been identified as key players in facilitating the much-needed energy transition [4–6]. These sources offer a cleaner and more sustainable alternative to traditional fossil fuels, which are major contributors to carbon emissions and environmental harm. The potential impact of renewable energy on our lives in the long term cannot be understated. By harnessing these natural resources, we have the opportunity to significantly reduce carbon emissions and mitigate the harmful effects of climate change. According to the International Energy Agency [7], utilizing renewable energy sources could lead to more than 80% of emissions reductions by 2030, primarily through electrification and decreasing CO_2 emissions from fossil fuel operations. Despite the clear benefits of renewable energy, its widespread adoption has been hindered by various challenges. Concerns regarding supply stability and the adequacy of technological infrastructure have led to hesitance in fully embracing renewable energy sources. However, with advancements in technology and increased awareness of the urgency of addressing climate change, the transition to renewable energy is becoming increasingly feasible and necessary for a sustainable future.

Electricity access is a vital enabling factor for human development, and the seventh United Nations Sustainable Development Goal (SDG) for 2030 acknowledges its significance [8]. However, one billion people lacked access to electricity around the world in 2017 which is 13% of the world population. Nearly half of these people live in sub-Saharan Africa, and a large majority of them live in rural areas [9]. Even if rural areas are sparsely populated, these communities have an equal right to equitable access for energy towards economic development [10]. To meet progress in SDG 7, it is necessary to consider sparsely populated communities. Thus, new approaches to the electricity system are required. To meet SDGs, increase in rural electrification is an important factor that requires thinking and planning judiciously the amount and the type of energy resources to use in the future. Moreover, equitable access for energy and an increase in consumption combined with pressing topics like global warming, availability of resources, and electricity costs are driving politicians and scientists around the world to find solutions to these impending threats [11]. One of the preferable solutions is to use grid extension in densely populated areas which is justifiable by the demand [12]. However, for sparsely populated regions, it is often unable to justify grid infrastructure extensions to those areas because of the economical distance constraint and low economic income potential for centralized electricity grid infrastructure [13]. In centralized electricity generation, electrical power transmission, and distribution mount up to 30% of the electricity costs. This share is the lowest for industrial consumers (medium and high voltage) and the highest cost for smaller consumers, such as residential or commercial buildings (low voltage electricity) [14, 15]. Inefficient grid growth rate coupled with the problem of grid quality and reliability, along with rapid population growth necessitates seeking options other than through utility grid extension depending on local renewable resource and capacity for rural electrification of rural community in many developing nations and entrench green economies in the process by decreasing CO_2 emission in power generation [16]. This has resulted in significant interest in renewable energy sources based off-grid energy generation especially for remote areas. Moreover the world is facing fossil fuel depletion, high energy costs, and increasing environmental concerns of conventional type energy technology [9]. According to the International Energy Agency, mini-grid and off-grid systems will provide electricity to 60% (18%) from stand-alone systems and 42% from mini-grids) of those gaining access to electricity in African rural areas by 2030 [17]. The term mini-grid or micro-grid has been defined in various sources. In practice, the literature uses the term “microgrid” to refer to systems in the order of tens to hundreds of kW. The term “mini-grid” has been associated with larger decentralized systems in the order of MW [18, 19]. In design and operation of hybrid renewable energy systems (HRES), optimal planning of the numerous components of energy resources is essential. Optimal sizing of each element is helping us to reduce system costs. Researchers have conducted numerous studies to explore the efficiency and effectiveness of planning on-grid and off-grid renewable energy systems. One key area of focus in these studies is determining the optimal mix of renewable energy sources, such as solar, wind, hydro, and biomass, to meet the energy demand of a specific location. Few of these studies are as follows.

In Abraha *et al.*’s study on the application of a hybrid system in three rural villages compared to a diesel-only system [20], it was found that the diesel-only power system was a more feasible option than implementing a solar–wind–diesel hybrid system. This conclusion was reached after evaluating factors such as cost-effectiveness, reliability, and sustainability. Bekele's study in Ethiopia used the HOMER optimization tool and sensitivity analysis to assess the feasibility of a Wind-PV hybrid system to electrify 200 families in rural areas [21]. The research highlighted that the generation cost from wind-PV systems was higher than the existing tariff in Ethiopia, indicating challenges in economic viability. Another study focused on rural household electrification in Ethiopia concluded that a wind–solar PV system was feasible for remote areas [22]. However, the research also pointed out the need for careful consideration of generation costs and tariffs to ensure financial sustainability [23]. Furthermore, a study presented a hybrid micro-hydro and wind power system for a rural area in Ethiopia with 660 households highlighted a levelized cost of electricity [24] of $0.112/kWh from the proposed system. This research emphasized the importance of evaluating the cost-effectiveness and reliability of hybrid systems in rural electrification projects [25]. Overall, these studies underscore the importance of considering various factors such as cost, reliability, and local conditions when designing and implementing hybrid power systems for rural electrification in developing countries like Ethiopia. Kebede [26] presented a step by step calculation approach to design a solar PV system for a modern typical home in Shewa Robit, Ethiopia, but using step by step calculation for designing of solar PV is time-consuming and may not obtain accurate results. Nigussie *et al.* [27] investigated the techno-economic and optimum configuration of a micro-hydro, PV, and Diesel Generator-battery hybrid power system for supplying electricity to Melkey Hera village in Western Ethiopia, which is remotely located from the national grid, using HOMER software. The three-year load projection was used as input by taking population growth of the community living in the village. Halabi *et al.* [28] analyzed the performance of a hybrid system based on economic and environmental constraints, besides emphasizing comparative cost analysis between different operational scenarios. HOMER software was used to model, simulate, and analyze hybrid systems and perform comparative cost analysis between different configurations including net present cost (NPC) and levelized cost of energy (LCOE). They concluded that a hybrid PV/diesel/battery system provided the best performance compared to all other scenarios and 100% RE system showed the best environmental characteristics with the highest costs. Ghaem Sigarchian *et al.* [29], proposed techno-economic optimized design of hybrid power system (PV/wind/battery and biogas engine as backup) for villages in Kenya and evaluate the performance of a system using HOMER software. HOMER is a powerful tool for the design of a hybrid system but lack of flexibility in altering the operating strategy and objective function are the limitations. Moreover, the sizing of the system components assumes many simplifications during the optimization process, which has a significant impact on the accuracy of results deduced from HOMER software. More so, the lowest price version of the HOMER software is not always the optimum solution, but it may be the best possible from the available possibilities entered for each component as a data. Overall, the literature indicates that off-grid hybrid renewable energy systems have the potential to play a significant role in rural electrification efforts in Ethiopia. Conducting techno-economic analyses can help policymakers and energy planners make informed decisions about the deployment of sustainable energy solutions to improve access to electricity and foster socioeconomic development in rural areas.

While there have been studies analyzing the techno-economic aspects of off-grid hybrid renewable energy systems for rural electrification in various regions, there is limited research specifically focusing on the Ethiopian context. Considering Ethiopia's unique geographical and socio-economic characteristics, such as its diverse topography and rapidly growing energy demand, there is a need for comprehensive studies that specifically assess the techno-economic viability and feasibility of implementing such systems in rural Ethiopian communities. Addressing this research gap can provide valuable insights into the challenges, opportunities, and optimal solutions for sustainable electrification in Ethiopia, ultimately contributing to the country's energy transition and overall development goals.

Figure 1 presents a typical model of Hybrid renewable energy system consisting of PV-wind- micro-hydro–diesel generator-battery systems. Adoption of HRES will encourage a sustainable future for all.

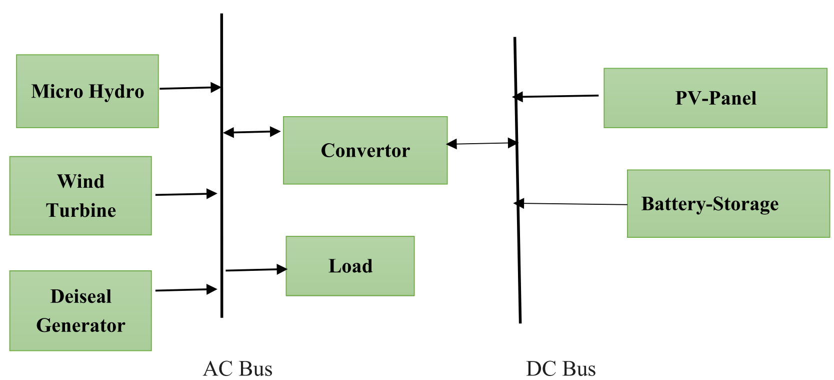

Figure 1. Block diagram of a typical off-grid Hybrid power plant.

2. Methodology

The study was carried out at three distinct locations in Ethiopia and culminated in a comprehensive analysis that encompassed several key stages. First, field visits were conducted to gather pertinent data concerning renewable energy resources. Subsequently, load analysis and load forecasting were performed for the chosen rural communities. Moreover, an estimation of renewable energy resources and an evaluation of cost optimization utilizing HOMER Pro software were carried out.

Table 1 and Figure 2 provide a visual representation of the geographical positions of the selected study locations. These stages collectively contributed to a thorough examination of the potential for utilizing renewable energy sources in the studied regions, aiming to enhance sustainable energy solutions for rural communities in Ethiopia.

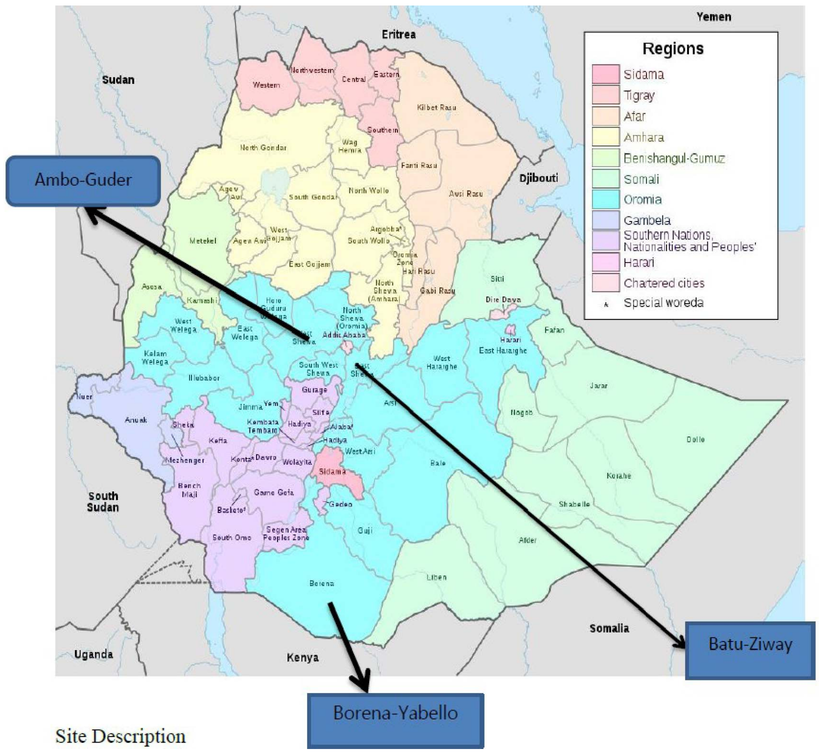

Figure 2. Selected study area.

Table 1. Study areas.

Places	Latitude	Longitude
Ambo-Guder	8°59′N	37.850°E
Ziway-Batu	7°56′N	38°43′E
Borena-Yabello	4°53′N	38°5′E

Based on the study protocol, the activities commenced with a comprehensive literature survey to gather existing knowledge on the subject. This was followed by the collection of primary and secondary data, with a focus on obtaining a wide range of information related to the selected site. Primary data were sourced from various governmental offices located near the chosen site. This data included key indicators such as the number of electrified individuals in the community, the trend of electricity usage within the community, and the income levels of both farmer and pastoral communities. Additionally, data regarding the characteristics and current conditions of run-off rivers, such as the maximum and minimum flow rates and available heads, were sourced from the Ministry of Water and Energy. Furthermore, information on daily solar radiation and monthly average wind speeds were obtained from the national meteorology agency and NASA, respectively. The study also involved performing economical and sensitivity analyses to draw research conclusions. The overall process of the study is visualized in Figure 3, providing a clear overview of the methodology followed in the research process and organized as follows: (1) Literature Review: We conducting a thorough review of existing literature to identify gaps in knowledge and establish a foundation for our research. (2) Load Analysis and Forecasting: Understanding the energy demands of the Determined Community to optimize energy distribution and usage. (3) Renewable Resource Energy Estimation: Assess the availability and potential of renewable energy sources in the community. (4) Designing Appropriate Hybrid System: We create a hybrid energy system that integrates various renewable sources and storage systems for increased efficiency. (5) Analysis with HOMER Pro Software: The latest version of HOMER Pro software (HOMER Pro Version 3.15 developed by the National Renewable Energy Laboratory in the United States.) was used for in-depth analysis of the hybrid system's performance. (6) Optimization based on Parameters: Evaluating the hybrid system based on key parameters like capital cost, cost of energy, unmet electric load, and capacity shortage to determine the most optimal solution. (7) Conclusion: Finally, the findings and implications of this study are summarized, including recommendations for future actions or improvements.

2.1. Load estimation

The load estimation for a kebele with 180 households in rural Ethiopia is a crucial step in ensuring adequate provision of electricity to meet the basic needs of the community members. The estimated electrical load includes various essential appliances such as energy-efficient Compact Fluorescent Lamps (CFLs), television, radio, water pump, flour mill, medical equipment, and stove [30]. In addition to the basic household needs, special consideration is given to the proposed site's primary school, milk processing facility, enjera mitad, and farmer training center. Accordingly, the estimated electrical load includes, three 11 W CFLs, out of which two for indoor lighting, and one for outdoor lighting, a 15 W radio receiver, a 70 W rated TV, the primary school consists of 8 classrooms and each classroom will be installed with three 11 W lamps. In addition, two 11 W lamps are assumed for office light and two another 11 W lamps are also considered for external lighting. Evening classes are conducted from 18:00 to 21:00. The school requires another 104 W load for office loads such as computer and printer to be operated between 9:00 and 4:00. One milk processing machine rated capacity of 4 kW, one flour mill with rated

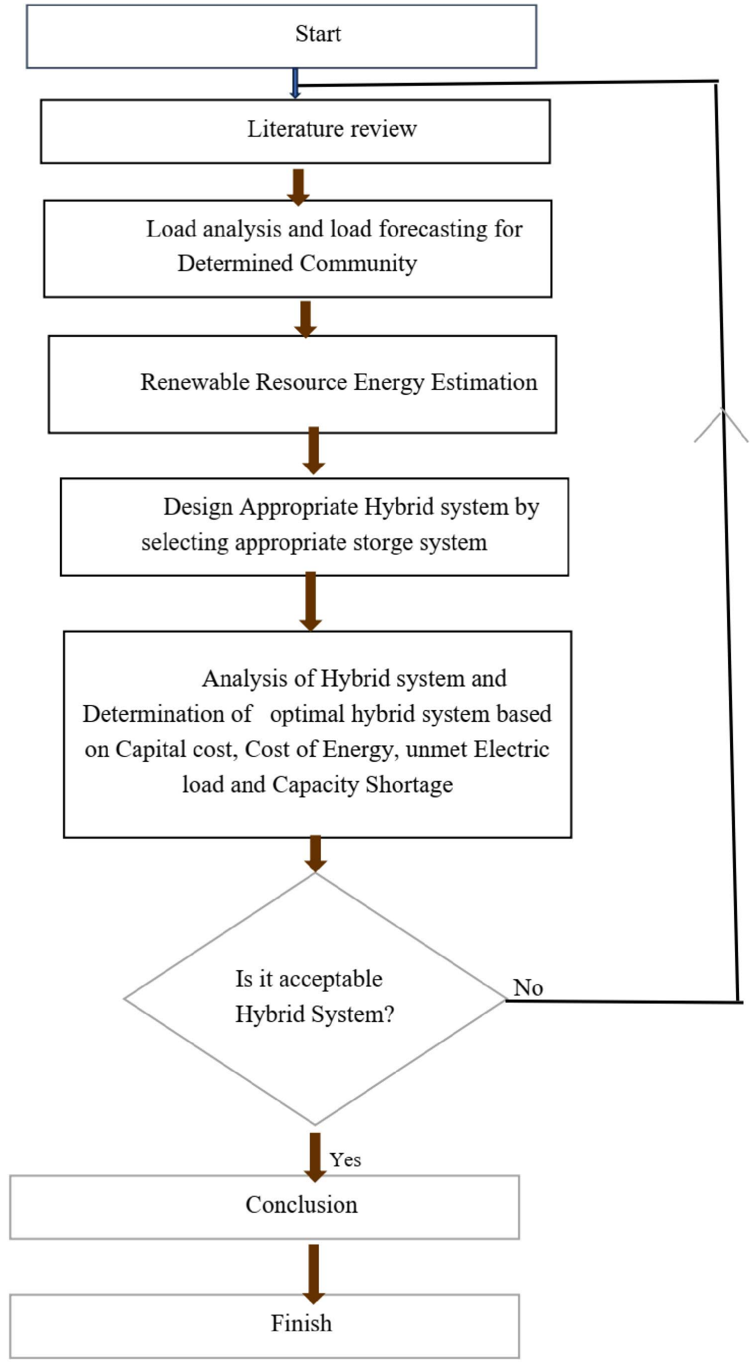

Figure 3. Study Protocol.

capacity of 7.5 kW with milling capacity are assumed to forecast the energy demand of the selected kebele. In reality, the demand is not constant and varies from time to time, day to day and season to season and working days to weekend and is explained separately. The summary of the estimated electrical load of selected kebele is shown in Table 2 and are classified as domestic load, public service and other [31]. Electric stove proportion is assumed to be 10% of the total householders. The electric stove operates for 12 h a day (07:00 to 19:00) and every 10% of householders will use electric stove for 5 h in this time period every day. The electric Mitad operates for 12 h a day (06:00 to 18:00) and 7% households are likely to use electric Mitad once a day for 2 h every three days, while fridge is expected to be used for 22 h per day.

Table 2. Primary load forecasting for previous five years.

1	2	3	4	5
443.3	465.5	488.8	513.2	538.9

2.2. Load forecasting

Load forecasting techniques are crucial for utilities and energy companies to accurately predict and plan for future energy demand. There are several techniques commonly used for load forecasting, each with its own strengths and limitations.

(1) Time series analysis: This technique involves analyzing historical load data to identify patterns and trends in energy consumption over time. By using mathematical models such as autoregressive integrated moving average (ARIMA) or exponential smoothing, time series analysis can provide reliable short-term load forecasts based on past consumption patterns.

(2) Artificial intelligence and machine learning: With advancements in AI and machine learning algorithms, utilities can now leverage predictive models to forecast energy demand more accurately. Techniques such as neural networks, support vector machines, and random forests can analyze complex data sets and identify nonlinear relationships to improve forecasting accuracy.

(3) Weather-based forecasting: Weather conditions have a significant impact on energy demand, especially in industries like heating, ventilation, and air conditioning (HVAC). By incorporating weather data into forecasting models, utilities can adjust for temperature fluctuations and seasonal variations to make more precise load forecasts.

(4) Econometric models: Econometric models combine economic indicators, demographic data, and other external factors to predict energy demand based on market trends and consumer behavior. By considering factors like GDP growth, population changes, and industrial production, utilities can create more comprehensive load forecasts that factor in macroeconomic influences.

(5) Hybrid forecasting approaches: Many utilities use a combination of techniques to improve the accuracy of load forecasts. For example, a hybrid model may integrate time series analysis with machine learning algorithms to leverage the strengths of both approaches and achieve better predictive performance. Overall,

choosing the most suitable load forecasting technique depends on the specific requirements of the utility, the availability of data, and the desired level of accuracy. By continuously refining and adapting forecasting methods, utilities can optimize resource allocation, reduce operational costs, and ensure reliable energy supply for consumers In load forecasting, the linear regression method stands out as a simple yet effective approach for predicting electrical loads in scenarios where historical data is not readily available [30].

While traditional methods rely on historical data to make accurate forecasts, linear regression can be adapted for non-historical load forecasting by leveraging a specific formula to derive the necessary input data [12]. This formula, denoted as

$$E_n = E_o \left(1 + \frac{r}{100}\right)^n \tag{1}$$

where, E_n electric load at the nth year, E_o is the current electric load and r is the annual electric load growth (5%).

For the purpose of this forecast, a five-year historical dataset has been used to make projections for the next twelve years. The initial current electric load value is set at 443.3607 kWhr/d, with an annual electric load growth rate of 5%. By plugging these values into the formula, the primary load forecasting results have been generated and are detailed in Table 2.

Mathematical expression for the straight line: -

$$y = a_o + a_1 x_i \tag{2}$$

where, a_o is the intercept and a_1 is the slope.

Minimizing the sum of errors squared at each point is accomplished as follows.

$$\sum_{i=1}^{n} (y_i - a_o - a_1 x_i)^2 = \text{minimum.} \tag{3}$$

Find a_o and a_1:

$$a_o \quad \frac{\left(\sum y_i\right)\left(\sum x_i^2\right) - \left(\sum x_i\right)\left(\sum x_i y_i\right)}{N\left(\sum x_i^2\right) - \left(\sum x_i\right)^2} \tag{4}$$

$$a_1 \quad \frac{N\left(\sum x_i y_i\right) - \left(\sum x_i\right)\left(\sum y_i\right)}{N\left(\sum x_i^2\right) - \left(\sum x_i\right)^2} \tag{5}$$

a_o	a_1	x_k	N
3.889378892	0.0487901657	14	3

Finally using the above equations, we determine the load of the selected rural community in 2034. The forecasting is then done by considering 2020 as a reference year. Therefore, the load is determined to be 965 kWh/d. Figure 4 shows the daily load profile and forecasted load.

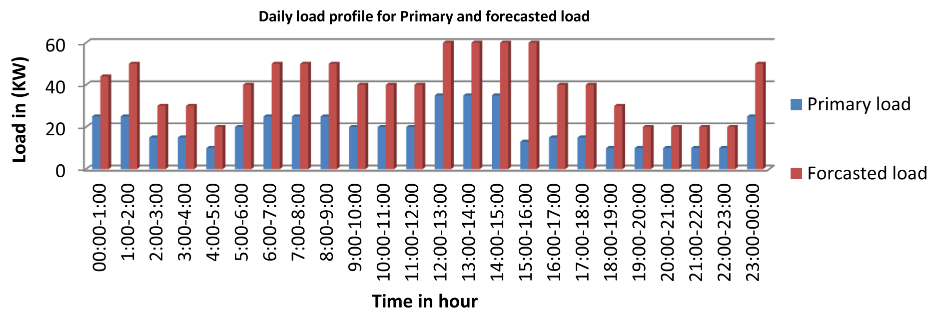

Figure 4. Primary and forecasted load.

2.2.1. Deferrable load

The concept of deferrable load is centered around tasks or appliances whose exact timing is not critical for their operation. This includes situations where the specific moment when the task is performed does not impact its overall effectiveness or outcome. A typical example of a deferrable load is a water pump used for various purposes such as drinking water and irrigation. In the case of water pumping for both drinking and irrigation needs, utilizing a water reservoir can be beneficial in ensuring a fair distribution of water to all members of the community within a designated area [12]. Different types of pumping equipment are available to facilitate the transfer of groundwater from lower to higher elevations. For this study, the Lorenz PS600 series with a power rating of 300 W and a pumping capacity of 0.67 liters per second (equivalent to 2400 liters per hour) was chosen [32]. Considering that the average depth in the selected area is around 15 m, it is essential to select equipment that can handle such requirements efficiently. Furthermore, it is important to note that in many developing countries, the average water consumption per household is approximately one hundred liters [33]. These factors contribute to the calculation of total forecasted deferrable loads, which are outlined in the Table 3. Additionally, agricultural loads for a two-hectare land are also taken into account within the study to provide a comprehensive analysis of the deferrable loads involved.

Table 3. Forecasted deferrable load of the study.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Load (kWh/day)	31.24	31.24	31.24	31.24	31.24	17	11.35	11.35	17	17	31.24	31.24

3. Resource estimation

3.1. Solar resource estimation

In order to compare the value of monthly solar radiation obtained from NASA, we used empirical formulas derived from general representation of sun and solar collectors' geometry as shown in Figure 5.

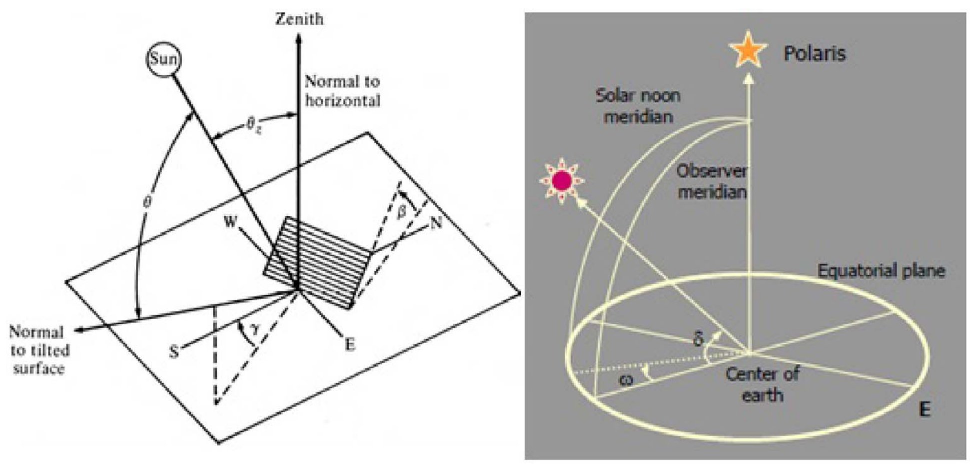

Figure 5. General representation of Sun and Solar collector [30].

Solar collector and Sun geometry general representation is modelled with latitude, longitude and collector surface angles. Based on these parameters, the monthly solar radiation was calculated using the relation between the following angles: the solar altitude, α_s, solar zenith angle, θ_z, the solar azimuth γ_s, latitude, ϕ, time of the year (declination angle, δ, time of the day (hour angle, ω) and angle of incident θ_i [34–37]. Table 5 depicts the 10-year average monthly solar radiation obtained from NASA and theoretically calculated results of the study sites is shown in Table 4 below.

The array power output can be obtained by the following relations when the effect of temperature on the PV array ignored; HOMER assumes that the temperature coefficient of power is zero [38]

$$P_{PV} = Y_{PV} f_{PV} \left[\frac{G_T}{G_{T,STC}} \right] \tag{6}$$

Y_{PV} is the rated capacity of PV array [kW], f_{PV} is the PV derating factor [%], G_T is the solar radiation incident on the PV array in the current step [kW/m²], $G_{T,\,STC}$ is the incident radiation at standard test conditions [kW].

Table 4. Monthly solar radiation of study sites.

Month	Ambo-Guder		Batu-Ziway		Yabello-Borena	
	H (kWh/m²/d)	NASA	H (kWh/m²/d)	NASA	H (kWh/m²/d)	NASA
Jan	5.87	5.76	5.93	6.13	5.52	6.21
Feb	6.35	5.35	5.01	5.2	5.54	5.51
Mar	6.57	6.09	6.25	6.24	5.68	5.59
Apr	6.63	5.67	5.97	5.87	5.99	5.45
May	6.56	5.42	5.5	5.7	4.92	5.09
Jun	6.18	5.43	5.27	5.37	5.05	4.57
Jul	5.59	4.28	4.3	4.4	4.15	4.23
Aug	5.22	4.13	4.9	4.7	4.07	4.53
Sep	6.48	5.11	5.02	5.05	5.92	5.36
Oct	6.40	6.26	5.92	6.11	5.23	5.13
Nov	5.96	5.73	6.23	6.14	6.09	6.05
Dec	5.71	5.28	6.02	6.03	6.33	6.26
Average	6.13	5.37	5.52	5.57	5.37	5.33

Table 5. Ten-year average wind speed for study areas at 50 m hub height.

Site/month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Yabello	6.6	6.87	6.86	5.77	5.85	6.16	6.63	6.9	6.37	5.86	6.32	6.16
Batu	10	11	11	10	9.5	9	9.5	9.5	8.5	10	12	12

3.2. Wind resource estimation

For this study we used data obtained from NASA. It is possible to extrapolate the selected wind turbine height using Equation (20) [37]. The value of Z_0 is taken as 0.1; Weibull parameters are estimated to be $K = 2$ and $c = 3.9$ m/s [37]. Table 5 summarizes 10-year wind speed at 50 m height.

$$v(z) \cdot \ln\left(\frac{Z_r}{Z_0}\right) = v(Z_r) \cdot \ln\left(\frac{Z}{Z_0}\right) \tag{7}$$

where, $Z_r \rightarrow$ Reference height (m), $Z \rightarrow$ Height where wind speed is to be determined (m), $Z_0 \rightarrow$ Measure of surface roughness (0.1 to 0.25 for crop land), $v(Z) \rightarrow$ Wind speed at height of Z m (m/s), $v(Z_r) \rightarrow$ Wind speed at the reference height (m/s).

The output power of wind turbine is dependent on the function of wind speed and its hub-height. Most studies use wind turbine power curve for a typical wind turbine generator as shown in Figure 6 [39].

The wind turbine starts generating power at cut-in speed and stops generating below cut-in speed and above cut-out speed. At rated speed which is given by manufacturer, wind turbine generator generates rated power. The equation shown

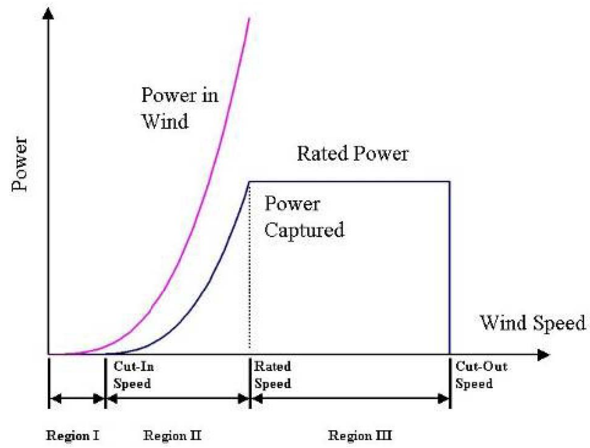

Figure 6. General power curve of wind turbine [39].

below is wind power generation equation which is used in most articles [39, 40]

$$P_W = \begin{cases} P_r \left(\dfrac{V(t)^2 - V_{cin^2}}{Vr^2 - V_{cin^2}} \right) & V_{cin} \le V(t) \le Vr \\ P_r & Vr \le V(t) \le V_{cout} \\ 0 & V(t) \le V_{cin} \text{ or } V(t) > V_{cout} \end{cases} \tag{8}$$

where: P_W is output power of wind turbine generator, P_r is rated power of wind generator, V_{cin} is the cut-in wind speed, and V_{cout} is the cut-out speed. Figure 7 illustrates the calculation of the output power of the turbine based on the specific speed input to the system. In Figure 7, the horizontal line indicates the hub-height wind speed, and the vertical line indicates the wind turbine power output.

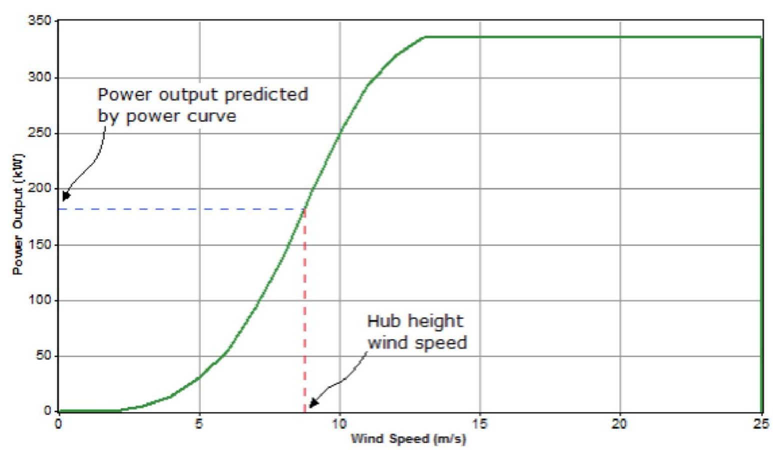

Figure 7. HOMER model wind speed versus turbine output power.

Based on the selected specific wind turbine characteristics, the calculated annual energy for each month of study area is shown in Table 6.

Table 6. Generated energy and speed of selected sites.

Month		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Yabelo	Speed (m/s)	6.6	6.87	6.86	5.77	5.85	6.16	6.63	6.9	6.37	5.86	6.32	6.16
	Energy (MWh)	2181	2202.26	2201.7	2034.2	2053.6	2118.4	2184	2203.8	2152.7	2056	2145	2118.4
Ziway	Speed (m/s)	7.9	7.35	7.43	6.91	6.06	6.03	6.29	6.36	5.49	7.89	8.075	8.21
	Energy (MWh)	2209.79	2187.9	2193	2138	1962	1954.21	2020	2036	1789	2209	2207	2203

3.3. Micro-hydro system and flow rate assessment for study area

A micro-hydropower system is a small system in the range of 5–100 kW [41]. The simplest micro-hydropower plant is based on a run-of-river design, which does not have water storage capability. It produces power only when water is running. Micro-hydropower is an interesting prospect for providing electricity for rural communities. The hydro turbine flow rates are the water quantity flowing through the hydro turbine.

HOMER software calculates this value in each time step using the following equations

$$Q_{turbine} = \begin{cases} 0 & \text{if } Q_{available} < Q_{min} \\ Q_{available} & \text{if } Q_{min} \leq Q_{available} \leq Q_{max} \\ Q_{max} & \text{if } Q_{available} > Q_{max} \end{cases} \tag{9}$$

where $Q_{available}$ = the flow rate available to the hydro turbine [m^3/s], Q_{min} = minimum flow rate of the hydro turbine [m^3/s], Q_{max} = maximum flow rate of the hydro turbine [m^3/s].

The nominal hydro power is the nominal power of the hydro system, or the power produced by the hydro turbine, given the available head and a stream flow equal to the design flow rate of the hydro turbine. The calculation of the nominal hydro power includes the efficiency of the hydro turbine, but not the pipe head loss. HOMER calculates the nominal Hydro power using the following equation

$$P_{hyd,nom} \frac{\eta_{hyd} \cdot \rho_{water} . g . h . Q_{desin}}{1000 \text{ W/kW}} \tag{10}$$

where: $P_{hyd,nom}$ = nominal power output of the hydro turbine [kW], η_{hyd} = hydro turbine efficiency [%], ρ_{water} = density of water [1000 kg/m^3], g = acceleration due to gravity [9.81 m/s^2], h = available head [m], Q_{desin} = the design flow rate of the hydro turbine [m^3/s].

Planning a micro-hydro generation requires the consideration of flow duration curve (FDC) [42]. Exact head and water flow rate are important parameters used to determine turbine size, speed and type for micro hydro power plant. To find the available water flow rate and maximum water flow capacity of the turbine, it is important to develop FDC for the selected run of river. We can get an idea of river's generation potential from average annual flow rate. It is possible to produce FDCs for specific periods of time and for specific years. FDC and 15-year average flow rate of the study area are shown on Figure 8 and Table 7, respectively.

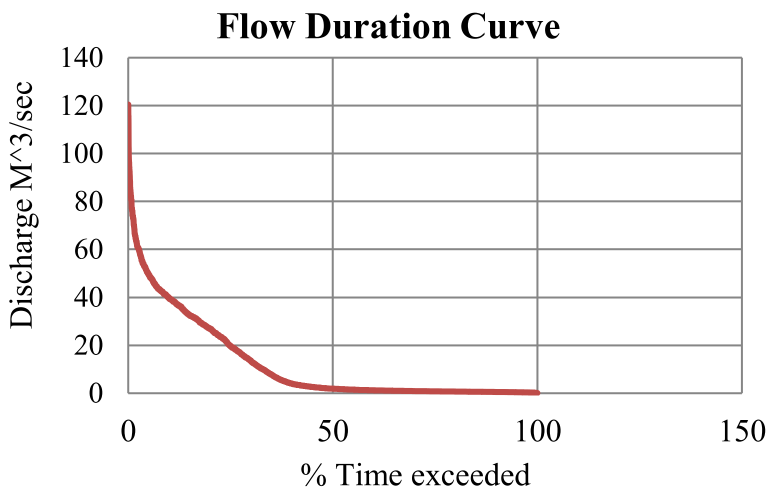

Figure 8. Flow duration curve for Guder River.

Table 7. 15-year average stream flow for Guder River.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Aver
Flow rate (l/s)	874	618	972.5	1162	1915	11,315	34,102	44,549	34,184	11,328	2828	1931.5	12,148.6

4. Results and discussion

To effectively simulate the overall system in HOMER software for recommending optimal hybrid renewable energy systems in various off-grid rural areas, a range of input parameters need to be considered across different resources. Table 8 outlines the key input parameters required for modeling wind turbines, hydropower systems, solar PV arrays, batteries, diesel generators, converters, primary loads, and deferrable loads within the software.

For wind turbines, parameters such as the type, cost (covering capital, replacement, and O&M costs), hub height, and lifetime are crucial inputs. Similarly,

Table 8. Input parameters.

Resource	Input parameter
Wind turbine	Type of wind turbine, cost of wind turbine (capital, replacement and O&M cost), hub height of wind turbine, life time of wind turbine)
Hydro power	Cost of Hydro turbine and generator (capital, replacement and O&M cost), design flow rate, flow rate ratio, efficiency of turbine and generator, head, life time of the system
Solar PV	Size, cost, slope angel, reflectance, derating factor and life time
Battery	Type of battery, cost of battery, size of battery string, life time
Convertor	Cost of convertors, efficiency of convertors, size of convertors and life time.
Primary loads	Annual hourly data
Deferrable loads	Monthly average daily load, Storage capacity

hydropower considerations include the cost of turbines and generators, design flow rate, flow rate ratio, efficiency of turbine and generator, head, and system lifetime. Solar PV inputs encompass size, cost, slope angle, reflectance, derating factor, and lifespan. Battery parameters include the type, cost, size of the battery string, and life expectancy. For converters, factors like cost, efficiency, size, and lifespan are essential inputs. Primary loads require annual hourly data, while deferrable loads involve monthly average daily load and storage capacity details. By utilizing these inputs, HOMER software generates graphical load profiles for both primary and deferrable loads for three distinct study areas and are shown in Figures 8, 9 and 10, respectively. To determine the optimal hybrid system for each locale, a selection criterion based on economic and technical parameters is employed. Economically, the system with the least net present cost (NPC) and the lowest cost of energy among all potential hybrid configurations is chosen. Sensitivity analysis is then conducted by varying prices of solar PV and diesel to select the most cost-effective hybrid system. By leveraging HOMER software and carefully considering these input parameters, the objective is to recommend efficient and sustainable hybrid renewable energy systems tailored to the specific needs of off-grid rural communities, ultimately fostering energy independence and resilience in these areas.

In addition to primary and deferrable loads, it is important to determine type, size, and costs of main hybrid components for all selected study areas. Table 9 describes main components of hybrid system considered in this study [43–45].

4.1. Results of site I (Ziway-Batu)

In site I (Ziway-Batu), a combination of solar and wind energy sources with a battery storage system has been implemented. The hybrid schematic HOMER model for site 1, as depicted in Figure 11, includes the necessary components for harnessing

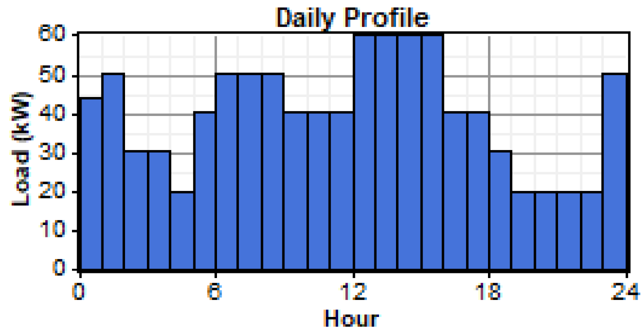

Figure 9. Primary load profile for 180 households.

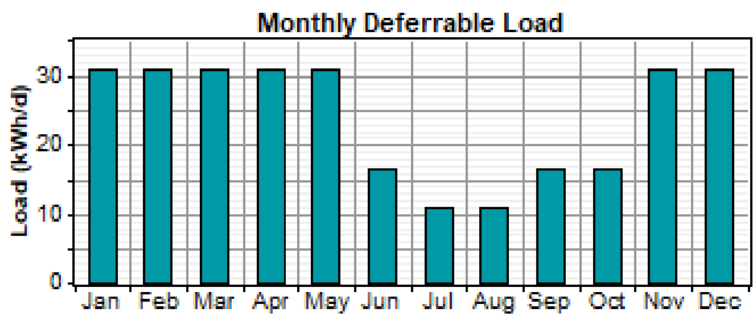

Figure 10. Monthly deferrable load for study community.

Table 9. Main components of hybrid system considered in this study.

Description	Size	Capital cost ($)	Replacement cost	Operation and maintenance cost ($)	Manufacturer
Solar PV	1 kW	370	370	42.6	Generic
Convertor (Cyb01000 A)	1 kW	400	400	0	Generic
Battery (1 kWh LA)	1 kWh	225	225	0	Generic
Diesel Gen (CAT-50)	40 kW	4930.8	4930.8	12+0.664/L	Caterpillar Inc
Wind turbine (XL10)	10 kW	12,500	12,500	500	Bergey Windpower Co
Micro-Hydro (hyd100)	100 kW	120,000	120,000	4400	Generic

renewable energy effectively. The site has a daily primary load of 964 kWh and a daily deferrable load of 24.24 kWh, with a peak capacity of 4.68 kW.

By integrating both solar and wind energy sources, the site aims to maximize energy generation and storage capacity while minimizing reliance on traditional non-renewable energy sources. The solar panels capture sunlight during the day to generate electricity, which can be stored in the battery system for later use. Similarly, the wind turbines harness wind energy to supplement the power generation process. The battery storage system plays a crucial role in ensuring a consistent power supply,

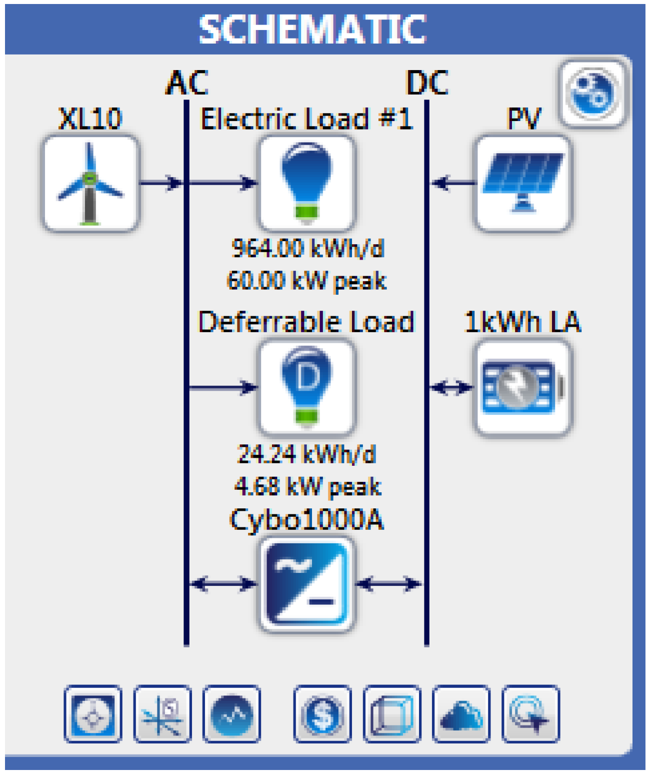

Figure 11. PV/wind/battery hybrid system for site I.

especially during periods when renewable energy generation is low. It allows excess energy to be stored and used when needed, thus reducing wastage and ensuring a more reliable energy supply. Overall, the integration of solar, wind, and battery storage technologies in site I demonstrates a sustainable approach to meeting energy needs while reducing carbon emissions and dependence on fossil fuels. The hybrid system's design and capacity have been optimized to meet the daily energy requirements of the site efficiently and effectively. Ten-year average wind speed data, solar radiation data and the characteristics of selected wind turbine generated for HOMER software are shown in Figures 12, 13 and 14, respectively.

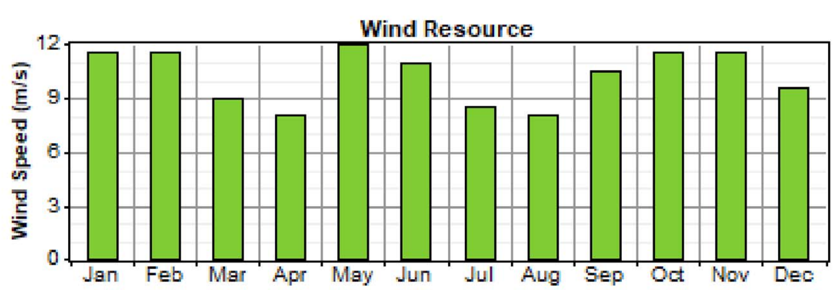

Figure 12. Wind speed data of Batu site.

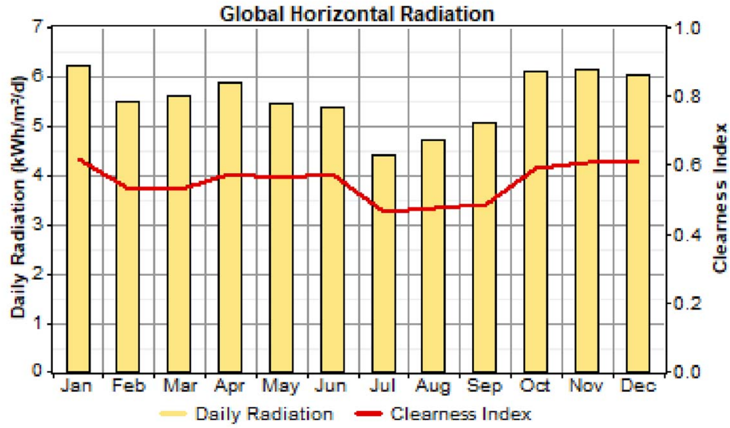

Figure 13. Solar radiation data of Batu site.

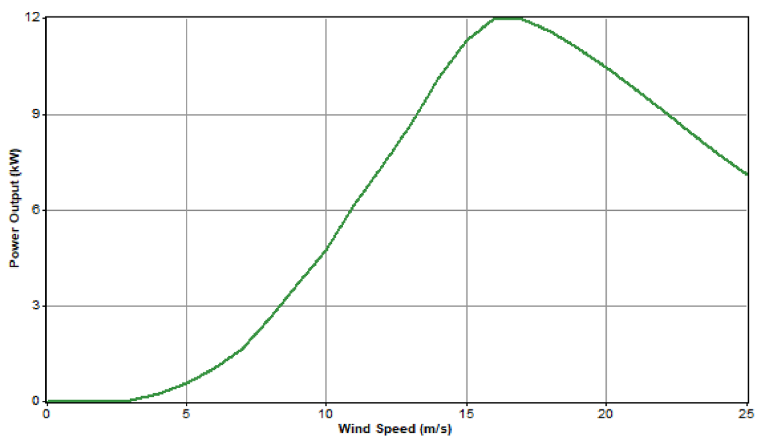

Figure 14. Power curve of the selected Wind Turbine.

When evaluating the results of this site, we considered the prices of the components as indicated in Table 10. The load flowing and cyclic charging dispatch strategy are both viable for this model. We analyzed the optimization results from both economic and technical perspectives. The system that ranks first in our assessment showed a surplus of 951 kWh of energy, achieving the lowest Cost of Energy (COE) at $0.112 per kWh. Furthermore, this system only has a 0.1% unmet load. At the other end of the spectrum, the system that ranked last displayed notable differences in technical and economic results. The additional COE for this last-ranking system amounted to $0.033 per kWh compared to the first-ranked system. The variations in the listed outputs of the overall optimization table, which ranged from $0.122 per kWh to $0.145 per kWh for a unity PV cost multiplier in the COE, further highlight the differences between the top-performing and lowest-ranking systems. The system that ranked first exhibited a surplus of 951,819 kWh of energy, with a COE of $0.122 per kWh and a minimal unmet load of 0.1%. In contrast, the last- ranked system had a surplus of just 420 kW of energy and a COE of $0.145 per kWh. Moreover, the annual monthly average electric production and overall system output of the site are illustrated in Figure 15 and detailed in Table 10. These findings provide a comprehensive overview of the performance and efficiency of the systems evaluated in this study.

Table 10. Overall results of site I.

System architecture					
PV	Wind turbine	Battery	Invertor	Rectifier	Dispatch
20 kW	30	720	80 kW	80 kW	LF
Annual electrical production (kWh/yr)					
PV-array	Wind turbine	Excess electricity	Unmet load	Capacity shortage	Renewable friction
35,395	1,305,115	951,819	173	249	100%
3%	97%	71%	0.1%	0.1%	
Annual electric consumption (kWh/yr)					
Primary loads		Deferrable load		Total	
351,687		8,666		360,353	
98%		2%		100%	
Cost summary					
Total net present cost		Levelized cost of energy		Operating cost	
$458,126		$0.122/kWh		$6,988/yr	

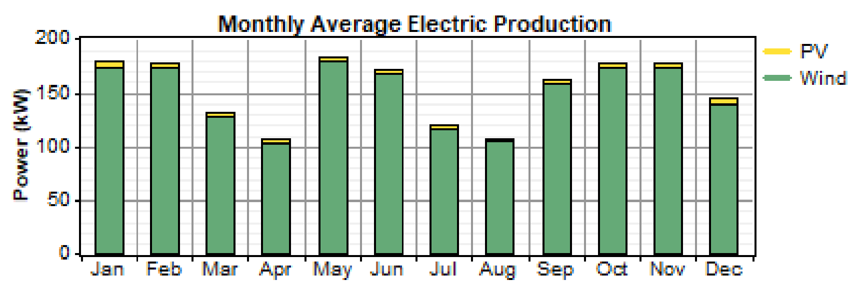

Figure 15. Monthly average electric production of site.

4.2. Results of site II (Yabello-Borena)

The results for site II (Yabello-Borena) show the detailed analysis and optimization of a hybrid solar, wind, diesel generator, and battery storage system. The HOMER model of the site included components for a 964 kWh daily primary load, 24.24 kWh daily deferrable load, and a peak capacity of 4.68 kW as shown in Figure 16.

The model also takes into account the ten-year average monthly wind speed, solar radiation, efficiency of the diesel generator, and the characteristics of the selected wind turbine for HOMER Pro software and are shown in Figures 17, 18, 19 and 20, respectively.

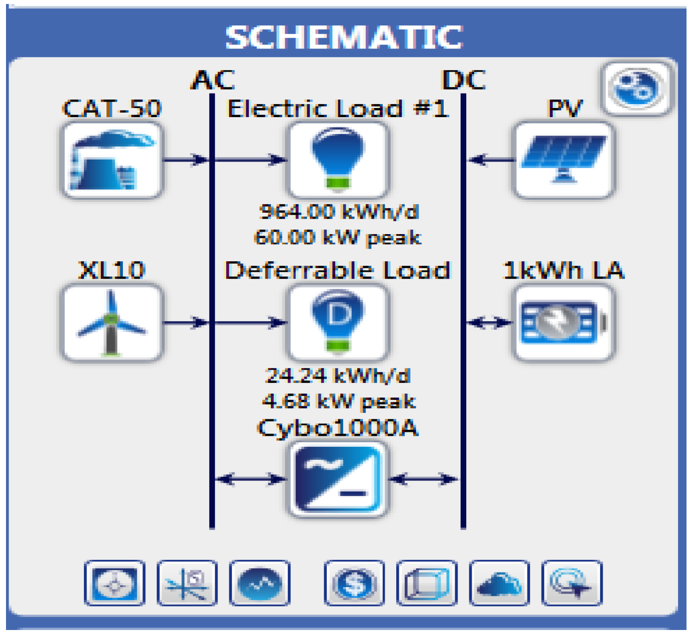

Figure 16. PV/wind/battery/diesel generator hybrid system.

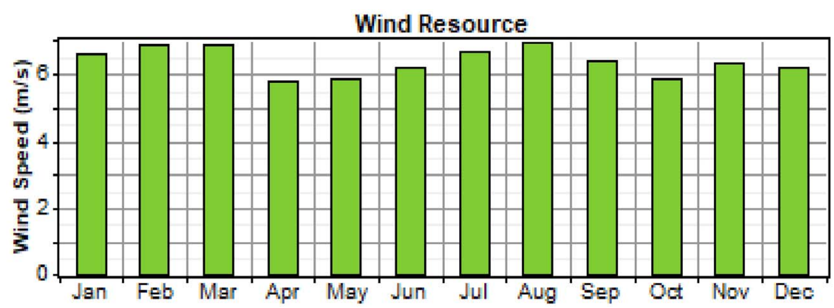

Figure 17. Wind speed data of Yabello site.

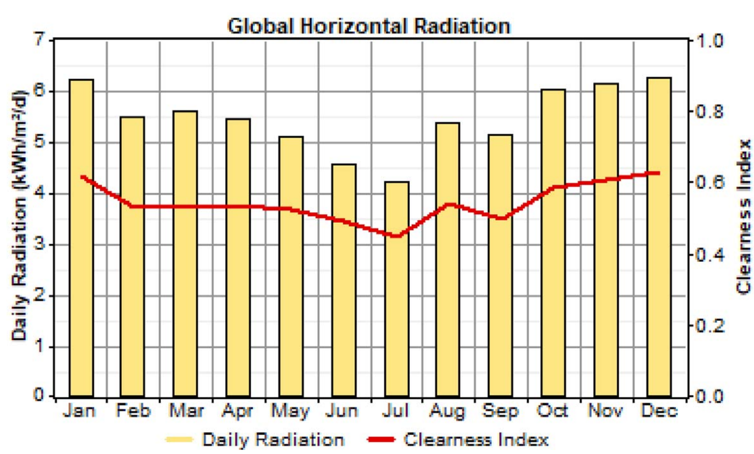

Figure 18. Solar radiation data of Yabello site.

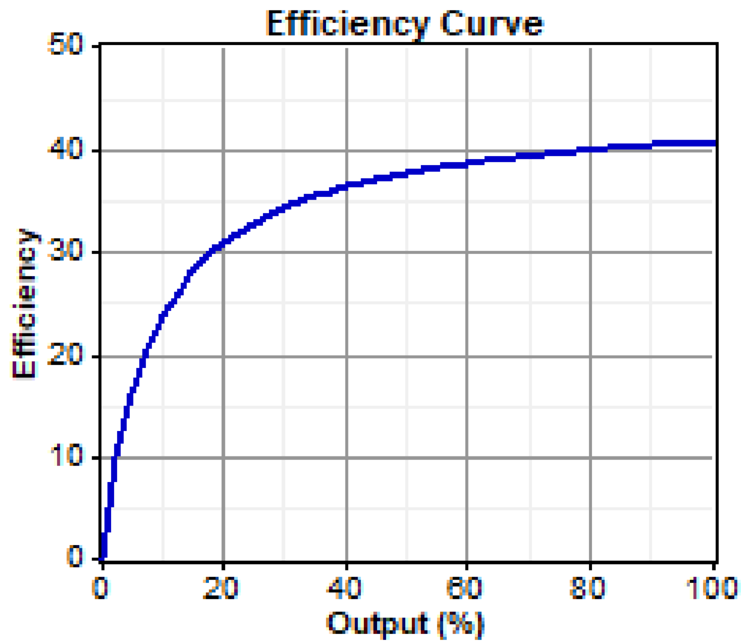

Figure 19. Generator efficiency curve of diesel.

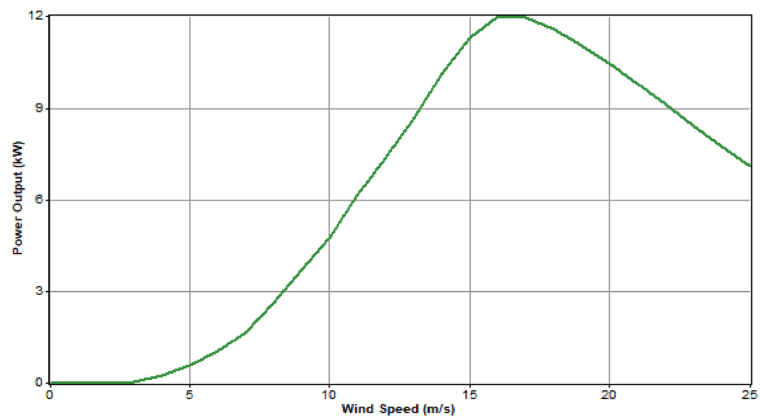

Figure 20. Power curve of selected wind turbine.

The analysis is based on current prices of PV, wind turbines, and fuel costs in the country. By varying the diesel price between $0.6/l and $0.8/l, the cost of energy ranged from $0.142/kWh to $0.156/kWh. Both load flowing and cyclic charging dispatch strategies were deemed feasible, with capacity shortage limited to a maximum of 0.1% and unmet load restricted to 0.1%. The optimization results indicate that all feasible combinations included a diesel generator, with a renewable fraction of 99%. The variations in PV/wind/diesel-generator/battery hybrid system configurations resulted in increasing total net present cost and cost of energy. The first optimal combination offered 322,657 kWh of excess energy with the lowest cost of energy at $0.142/kWh and 0.1% unmet load. The final combination had 620 kWh per year of excess energy with an additional cost of energy of $0.023/kWh and an unmet load of 0.1%. In cases where the diesel price changed to $0.8 and $1.2, the cost of energy adjusted to $0.138/kWh and $0.156/kWh, respectively. The annual monthly average electric production and overall system output for the site are illustrated in Figure 21 and Table 11, providing further insights into the performance of the hybrid system in site II.

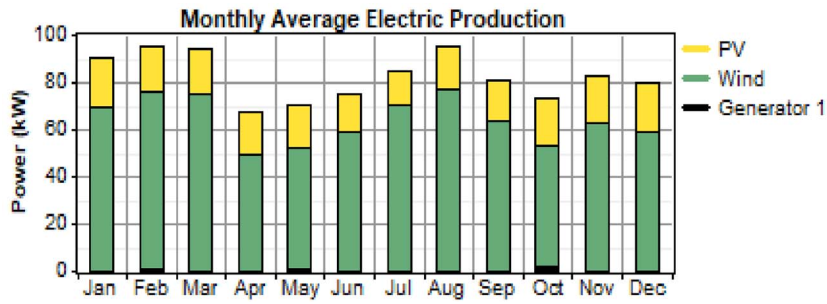

Figure 21. Monthly average electric production of site II.

Table 11. Overall system results for site II.

System architecture					
PV 100 kW	Wind turbine 30	Battery 720	Invertor 80 kW	Rectifier 80 kW	Dispatch LF
Generator 40 kW					
Annual electrical production (kWh/yr)					
PV-array	Wind turbine	Excess electricity	Unmet load	Capacity shortage	Renewable friction
163,328 22%	556,718 77%	322,657 71%	110 0.1%	308 0.1%	99%
Generator 2906 (1%)					
Annual electric consumption (kWh/yr)					
Primary loads 351,687 98%		Deferrable load 8,666 2%		Total 360,353 100%	
Cost summary					
Total net present cost $722,390		Levelized cost of energy $0.142/kWh		Operating cost $13534	

4.3. Results of site III (Guder-Ambo)

In the evaluation of site III (Guder-Ambo), a hybrid energy system incorporating solar, micro-hydro, and battery storage was considered. The HOMER model of this site, illustrated in Figure 22, includes all the mentioned components and catered to a daily primary load of 964 kWh along with a 24.24 kWh daily deferrable load, supported by a 4.68 kW peak capacity.

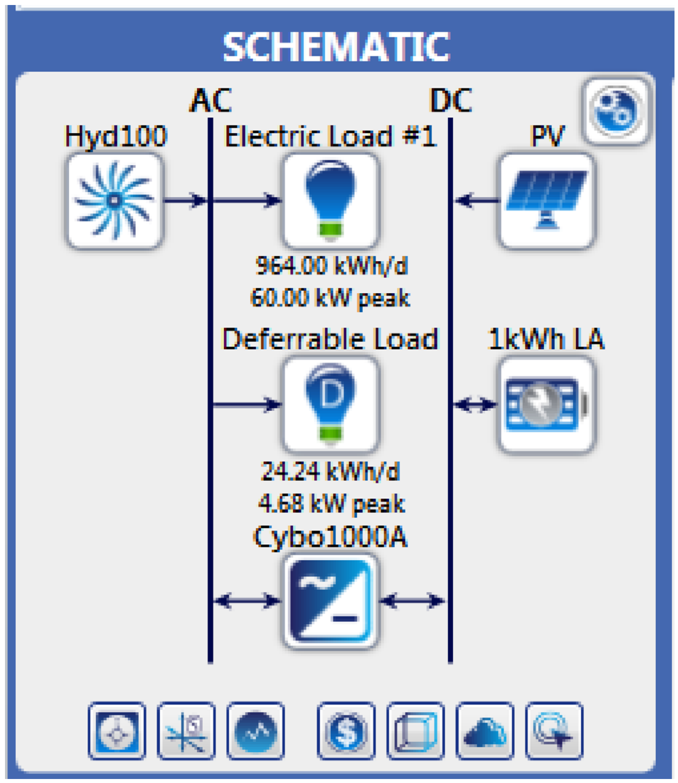

Figure 22. PV/micro-hydro/battery/hybrid system.

Ten-year average solar radiation data of the site and 15 years flow rate of Guder River generated for HOMER Pro software are shown in Figures 23 and 24, respectively.

Economic and technical assessments were conducted for this setup. The cost of energy for site III was within the range of \$0.043/kWh to \$0.0513/kWh. Both load flowing and Cyclic Charging dispatch strategies were found to be feasible without any capacity shortage or unmet load issues. An evaluation was carried out for a PV/micro-hydro/ battery hybrid system, showcasing an increase in total net present cost and cost of energy (COE). The optimization table showed different outputs for varying PV cost multipliers in the COE range. The system ranking first exhibited a surplus of 202,408 kWh of excess energy, boasting the lowest COE at \$0.043/kWh

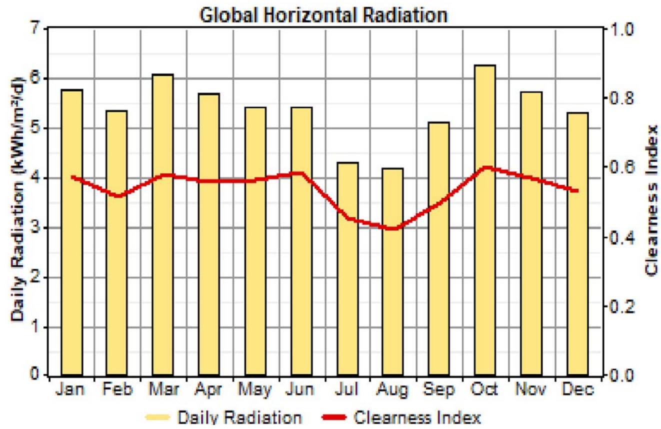

Figure 23. Solar radiation data of Guder site.

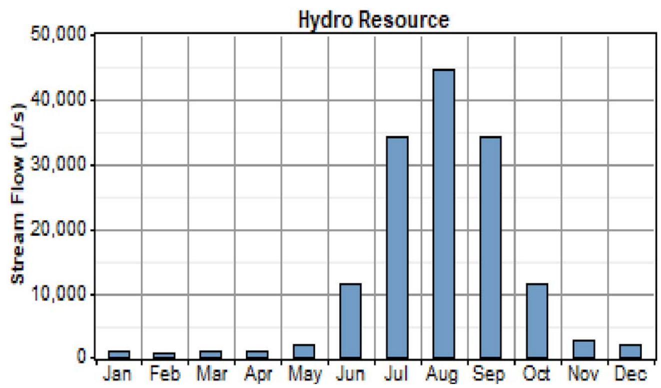

Figure 24. Flow rate data of Guder River.

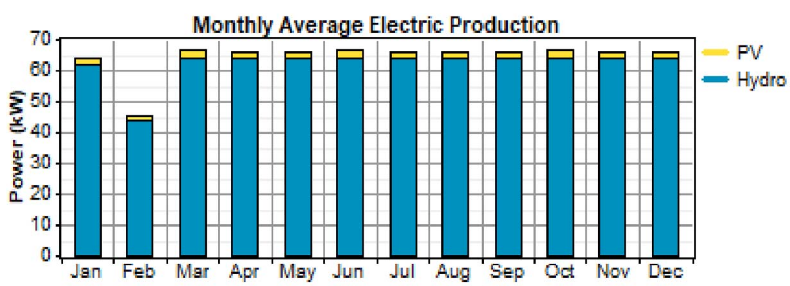

Figure 25. Monthly average electric production of site III.

and no unmet load. Conversely, the last- ranked system generated 108,314 kWh excess energy annually with a 0% unmet load, but at an additional COE of $0.0083/kWh. Additional details on annual monthly average electric production and overall system performance can be referenced for further insights shown in Table 12 and Figure 25.

Table 12. Overall system results for site III.

System architecture					
PV	Micro-hydro	Battery	Invertor	Rectifier	Dispatch
10 kW	50.4 kW	60	20 kW	20 kW	LF
Annual electrical production (kWh/yr)					
PV-array	Micro-hydro	Excess electricity	Unmet load	Capacity shortage	Renewable friction
16,329	547,877	202,408	0	246	100%
3%	97%	71%	0%	0.01%	
Annual electric consumption (kWh/yr)					
Primary loads		Deferrable load		Total	
351,687		8,666		360,353	
98%		2%		100%	
Cost summary					
Total net present cost		Levelized cost of energy		Operating cost	
$216,414		$0.043/kWh		$74,598/yr	

4.4. Comparison of study results with other studies

Table 13 elucidates the results of this study in comparison with findings from other studies in various areas such as power consumption, COE, Net Present Cost (NPC), Operation and Maintenance (O&M) costs, and electricity production. This comparison highlights the wide range of outcomes observed in the studies, reflecting the variability in renewable energy system performance. An important insight from this comparison is the significant difference in COE values among the studies. For example, in the study by Benti [46], the highest COE of 0.415$/kWh was recorded, mainly due to the limited utilization of renewable energy sources in their system design. This disparity in COE values emphasizes the impact of design choices and resource allocation on the overall cost efficiency of renewable energy systems. Furthermore, it is noted that other research studies exhibited shortcomings across dimensions such as shortage capacity, unmet load, renewable friction, and electric production. These deficiencies indicate areas where improvements are needed to enhance the performance and effectiveness of renewable energy systems and to address the challenges by focusing on optimizing system design, resource utilization, and operational efficiency, to achieve more sustainable and cost-effective renewable energy solutions.

Table 13. Comparison of study results with other studies.

Authors	Key objective	Country	Hybrid system	COE/kWh
Bekele & Palm [22]	Investigate the possibility of supplying electricity from a solar–wind hybrid system to a remotely located model community detached from the main electricity grid in Ethiopia	Ethiopia	Diesel/battery/ converter	0.322
Benti [46]	Assesses the suitability of stand-alone wind–solar PV hybrid power for Debmel village which is detached off the main grid line	Ethiopia	PV/diesel/battery/ converter	0.415
Kiros *et al.* [47]	A stand-alone hybrid system for the remote area of Ethiopia	Ethiopia	PV/wind/diesel/ battery/converter	0.402
Gebrehiwot *et al.* [48]	Assesses the potential of a hybrid system to electrify a remote rural village in Ethiopia	Ethiopia	PV/wind/battery/ diesel	0.207
Oladigbolu *et al.* [49]	Examines the potential application of hybrid solar PV/hydro/diesel/battery systems to provide off-grid electrification to a typical Nigerian rural village	Nigeria	PV/hydro/diesel/ battery	0.112
Benti *et al.* [50]	Techno-economic analysis of solar energy system for electrification of rural school in Southern Ethiopia	Ethiopia	PV/Battery/diesel	0.254

5. Conclusion

In conclusion, the study on planning an off-grid generation system using renewable energy sources for electrification in Ethiopian rural communities has shown promising results. By analyzing three different sites with varying renewable energy potential, micro-hydropower, wind energy, and solar energy, by utilizing HOMER Pro software for optimization, it was determined that a hybrid system of PV/Micro-Hydro/battery was the most cost-efficient for sites with good flow rates. Furthermore, in areas with sufficient wind speeds and solar radiation, it was possible to generate electricity at an affordable price without the need for additional diesel generators. For regions lacking adequate wind speeds, a hybrid system including a diesel generator, wind turbine, and solar PV can still provide a viable energy solution, especially when considering the current electrical prices in the country. This study highlights the importance of considering the seasonal nature of renewable resources by integrating wind and solar energies with a diesel generator and battery backup system for enhanced reliability in rural electrification. Ultimately, countries like Ethiopia, blessed with abundant renewable energy resources, have the opportunity to leverage these sources and bring electricity to remote communities that are beyond the reach of the national grid.

Author's contribution

Mekonnen, Abiy: Conceptualization, Data curation, Formal analysis, Investigation and Software, Methodology, Writing – original draft, Writing – review & editing; **Hiremath, Ravikumar:** Supervision, Validation, Visualization; **Shiferaw, Dereje:** Project administration, Resources.

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Ethical statement

Not applicable.

Data availability

Source data is not available for this article.

Conflicts of interest

The authors declare no conflict of interest

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